

1 Research The Invisible Engine & Digital Immune System

Role: Agentic AI Orchestrator & AI Security Researcher

Company: Tech Concept Lab (Blue Science Park)

Location: Karlskrona, Sweden

Timeline: Jan 2026 – Present

Executive Summary

"Moving AI from Generation to Action."

While most of the industry focuses on "Generative AI" (chatbots that write text), this project focuses on **"Agentic AI"**—autonomous systems capable of reasoning, planning, and executing complex multi-step business processes without human intervention.

Concurrently, this role involves pioneering **"Adaptive Resilience"** in cybersecurity—designing AI infrastructures that can detect cognitive attacks (like prompt injection) and self-repair in real-time.

System Architecture: "The Invisible Engine"

The "Invisible Engine" is a headless, autonomous orchestration layer designed to integrate seamlessly into enterprise operations. It does not wait for prompts; it acts on triggers.

1. Core Logic: Agentic Reasoning Pipelines

- **Frameworks:** LangChain, CrewAI.
- **Cognitive Architecture:** Implemented **ReAct (Reason + Act)** and **Chain-of-Thought (CoT)** prompting patterns to allow agents to break down complex objectives into executable sub-tasks.
- **Multi-Agent Collaboration:** Designed a "Crew" structure where specialized agents (e.g., a "Researcher," a "Strategist," and an "Executor") collaborate.
 - *Example:* One agent retrieves live market data, another analyzes it against internal KPIs, and a third triggers a specific API action based on the result.

2. Execution Layer: Workflow Automation

- **Orchestration Engine:** n8n (node-based workflow automation).
- **Integration Strategy:** The AI is not a silo. It is connected via API to CRM systems, Email servers, SQL databases, and Vector Stores.

- **Deterministic vs. Probabilistic:** Engineered a hybrid architecture where the AI provides the *reasoning* (Probabilistic), but the execution happens through rigid, code-based workflows (Deterministic) to ensure reliability and safety.

3. Reliability & Memory

- **Self-Correction Loops:** If an agent attempts an API call that fails, it analyzes the error message, refactors its payload, and retries automatically.
- **Long-Term Memory:** Utilization of Vector Databases (e.g., Pinecone/ChromaDB) to give agents "episodic memory," allowing them to recall past interactions and improve performance over time.



R&D Pillar: The Digital Immune System

This is the cutting-edge research component focused on **AI Security Architecture**.

Objective: "Adaptive Resilience"

To build an infrastructure that treats cyber threats not as static events, but as biological viruses that the system must develop immunity against.

Key Components:

- **Adversarial Simulation (Red Teaming):**
 - Deployed "Red Team" AI agents tasked with continuously attacking the main system using **Prompt Injection**, **Jailbreaking**, and **Cognitive Exploits**.
- **The "Sentinel" (Anomaly Detection):**
 - A real-time monitoring layer that analyzes input semantics. If an input deviates from known safe patterns (e.g., an attempt to override system instructions), it is flagged before reaching the core model.
- **Self-Healing Mechanism:**
 - Upon detecting a breach or a successful injection, the system isolates the session and automatically updates its "System Prompt" or guardrail configuration to block that specific semantic vector in the future.
- **Research Output:**
 - Currently documenting these findings in a white paper titled **"Adaptive Resilience: Engineering Self-Healing AI Architectures for Enterprise Security."**



Tech Stack & Tools

- **Orchestration:** LangChain, CrewAI, AutoGen.
- **Workflow Automation:** n8n (Self-hosted).
- **LLMs:** GPT-4o, Claude 3.5 Sonnet (for reasoning), Llama 3 (for local privacy-focused tasks).
- **Vector Database:** ChromaDB / Pinecone.
- **Backend:** Python, FastAPI (for custom agent tools).

- **Security:** Lakera Guard (conceptually), Prompt Engineering Defense patterns.



Strategic Impact

1. **Operational Efficiency:** transitioning manual enterprise workflows into "set-and-forget" autonomous loops.
2. **Security Leadership:** Defining new standards for how organizations protect LLMs from cognitive threats.
3. **Innovation:** Proving that AI can be a safe, reliable *actor* in the business world, not just a content generator.